

Agenda:

1. Module Introduction - [LINK](#)
2. Bitwise Operations [AND OR XOR & NOT]
3. Basic AND OR & XOR Properties
4. Left Shift & Right Shift
5. Check Bit / Set Bit / Flip Bit / Unset Bit
6. Find the min XOR pair in array

Class Rules:

1. Make sure to come on time (+5 min buffer in case of emergency). Late entry will not be permitted.
2. Mobile phones should be in pockets/bags in silent mode.
3. Earphones, headphones, etc. should be in bags.

Bits: 0 => Off / Unset Bit / False

1 => On / Set Bit / True

Decimal of **10100** = 20

Binary of **45** = 101101

16 8 4 2 1
1 0 1 0 0

R
↓

16 + 4 = 20

2	45	1
2	22	0
2	11	1
2	5	1
2	2	0
2	1	1
	0	

Bitwise Operations [AND, OR, XOR & NOT]

		AND	OR	XOR (exclusive OR)	NOT
a	b	a&b	a b	a^b	~a
0	0	0	0	0	1
0	1	0	1	1	1
1	0	0	1	1	0
1	1	1	1	0	0

0+0=0 1+0=1
0+1=1 1+1=10

20 & 45 = ?

20 | 45 = ?

20 → 0 1 0 1 0 0

45 → 1 0 1 1 0 1

0 0 0 1 0 0 → 4

20 → 0 1 0 1 0 0

45 → 1 0 1 1 0 1

1 1 1 1 0 1

32 16 8 4 2 1

61

20 ^ 45 = ?

20 → 0 1 0 1 0 0

45 → 1 0 1 1 0 1

1 1 1 0 0 1 → 57

Basic Properties

1. Even / Odd Number →

7 → 1 1 1 10 → 1 0 1 0

7 → 1 1 1 10 → 1 0 1 0

9 → 1 0 0 1 8 → 1 0 0 0

9 → 1 0 0 1 8 → 1 0 0 0

11 → 1011

11 → 1 0 1 1 10 → 1 0 1 0

11 → 1 0 1 1 10 → 1 0 1 0

$\& \quad 0 \quad 0 \quad 0 \quad 1 \quad \quad \quad \& \quad 0 \quad 0 \quad 0 \quad 1$

$\& \quad 0 \quad 0 \quad 0 \quad 1 \quad \quad \quad \& \quad 0 \quad 0 \quad 0 \quad 1$

0 0 0 1

0 0 0 1

$N \% 2 \rightarrow 0 \Rightarrow \text{even}$

$$(N \& 1) \nabla \rightarrow 1 \Rightarrow \text{odd}$$

$N \% 2 \swarrow$ $0 \Rightarrow \text{even}$

2. $A \& 0 \rightarrow 0$

3. $A \& A \rightarrow A$

2. $A \& 0 \rightarrow \underline{0}$ 3. $A \& A \rightarrow \underline{A}$

5 → 10 1

5 → 10 1

$0 \rightarrow \underline{000}$
 $5 \rightarrow \underline{101}$

$0 \rightarrow \underline{000}$
 $5 \rightarrow \underline{101}$

$000 \rightarrow 0$ $101 \rightarrow 5$

$000 \rightarrow 0$ $101 \rightarrow 5$

4. $A | 0 \rightarrow \underline{A}$

5 $\rightarrow$ 1 0 1

0 $\rightarrow$ 0 0 0

1 0 1 $\rightarrow$ 5

5. $A | A \rightarrow \underline{A}$

5 $\rightarrow$ 1 0 1

5 $\rightarrow$ 1 0 1

1 0 1 $\rightarrow$ 5

6. $A \wedge 0 \rightarrow \underline{A}$

5 $\rightarrow$ 1 0 1

0 $\rightarrow$ 0 0 0

1 0 1 $\rightarrow$ 5

7. $A \wedge A \rightarrow \underline{0}$

5 $\rightarrow$ 1 0 1

5 $\rightarrow$ 1 0 1

0 0 0 $\rightarrow$ 0

Cumulative Property $\rightarrow$

$A \& B = B \& A$ // Bitwise AND

$A | B = B | A$ // Bitwise OR

$A \wedge B = B \wedge A$ // Bitwise XOR

---1---0---

Associative Property $\rightarrow$

$(A \& B) \& C = A \& (B \& C)$ // Bitwise AND

$(A | B) | C = A | (B | C)$ // Bitwise OR

$(A \wedge B) \wedge C = A \wedge (B \wedge C)$ // Bitwise XOR

Left Shift Operator (<<)

8 bit +ve system

int $\rightarrow$ 32 bits $[-2^{31} \quad (2^{31}-1)]$
 $\sim [2 \times 10^9 \quad 2 \times 10^9]$

	128	64	32	16	8	4	2	1	
a =	0	0	0	0	1	0	1	0	$\rightarrow 10$
a << 1	0	0	0	1	0	1	0	0	$\rightarrow 20$
a << 2	0	0	1	0	1	0	0	0	$\rightarrow 40$
a << 3	0	1	0	1	0	0	0	0	$\rightarrow 80$
a << 4	1	0	1	0	0	0	0	0	$\rightarrow 160$
a << 5	0	1	0	0	0	0	0	0	$\rightarrow 320$

$(1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1)_2 \rightarrow 2^7 + 2^6 + 2^5 \dots 2^0 = 2^8 - 1 = 255$

$$A \ll 1 = A * 2$$

$$A \ll n = A * 2 * 2 * 2 \dots n \text{ times} = A * 2^n$$

int $\frac{0}{31} \ \frac{1}{30} \ \frac{0}{29} \dots \frac{0}{0} \rightarrow 2^{30}$

$$2^{30} \ll 1 = 2^{31}$$

$$(1 \ 0 \ 0 \dots 0)_2 \rightarrow -2^{31}$$

Right Shift Operator (>>)

	128	64	32	16	8	4	2	1 discarded
a =	0	0	0	0	1	0	1	0 → 10
a >> 1	0	0	0	0	0	1	0	1 → 5
a >> 2	0	0	0	0	0	0	1	0 → 2
a >> 3	0	0	0	0	0	0	0	1 → 1
a >> 4	0	0	0	0	0	0	0	0 → 0

$$A \gg 1 = A / 2$$

$$A \gg n = A / 2^n \quad \text{if } (2^n > A) \Rightarrow A / 2^n = 0$$

N = 45 →

	5	4	3	2	1	0
	1	0	1	1	0	1

0 0 ... 1 0

Set Bit i → $N | (1 \ll i)$

Flip Bit i → $N \wedge (1 \ll i)$

Check Bit i → $N \& (1 \ll i)$

0 → unset
1 → set

Unset ith Bit → check
+ flip

N = 5
(1 << 2)

	i	2	1	0
		1	0	1
		1	0	0
		1	0	0

31	30		4	3	2	1	0
1	1	...	1	0	1	1	1

$\sim (1 \ll i)$

Q: Given an integer array **A[]** of **N** integers, find the pair of integers in the array which have

minimum XOR value. Report the minimum XOR value.

same bits

$$\min \underline{A[i] \wedge A[j]} \text{ s.t. } (i \neq j)$$

Example:

A = [0, 2, 5, 7]

$$0 \wedge 2 = 2$$

$$0 \wedge 5 = 5$$

$$0 \wedge 7 = 7$$

$$2 \wedge 5 = 7$$

$$2 \wedge 7 = 5$$

$$5 \wedge 7 = 2$$

$$\text{ans} = \underline{2}$$

0 1 2 3 4 5

A = [3, 10, 12, 4, 7, 14]

	3	2	1	0
3	0	0	1	1
10	1	0	1	0
12	1	1	0	0
4	0	1	0	0
7	0	1	1	1
14	1	1	1	0

$$\underline{0010} \rightarrow \underline{2}$$

$$\begin{matrix} 10000 \\ a \\ 16 \end{matrix}$$

$$\begin{matrix} 01111 \\ b \\ 15 \end{matrix}$$

>

3	0	0	1	1
4	0	1	0	0
7	0	1	1	1

small $\begin{matrix} \nearrow 3 \text{ small} \\ \searrow 4, 7 \text{ large} \end{matrix}$

10	1	0	1	0
12	1	1	0	0
14	1	1	1	0

large $\begin{matrix} \nearrow \text{small} \\ \searrow \text{large} \end{matrix}$

smallest

largest

$$\text{sort}(A) \rightarrow \text{ans} = \min \bigvee_i (A[i] \wedge A[i+1])$$

$$TC = \underline{O(N \log(N))}$$

$$\text{ans} = A[0] \wedge A[1]$$

for $i \rightarrow 1$ to $(N-2)$

$$\text{ans} = \max(\text{ans}, A[i] \wedge A[i+1])$$

return ans